

AI Bias & Fairness Audit Report

Model: Demo Loan Model (v1.0)
Date: 2026-05-05 14:39:47.037179
Test Cases: 400

Overall Fairness Score: 0.97 — FAIR

Dimension	Score	DI Ratio	Status
Gender	1.00	1.00	✓ Fair
Caste	0.99	0.99	✓ Fair
Language	1.00	1.00	✓ Fair
Region	0.88	1.12	✓ Fair

Bias Analysis

Gender (Severity: LOW)

Minimal bias detected in the Gender dimension. The fairness score of 1.00 indicates the model treats demographic variants largely equally, with only 15.0% average difference. The Disparate Impact Ratio of 1.00 meets the 0.80 threshold.

Root Cause: Minimal gender bias detected. The model treats gender-variant inputs largely equally with minor statistical variation.

Caste (Severity: LOW)

Minimal bias detected in the Caste/Identity dimension. The fairness score of 0.99 indicates the model treats demographic variants largely equally, with only 20.0% average difference. The Disparate Impact Ratio of 0.99 meets the 0.80 threshold.

Root Cause: Low caste-based disparity. Name-based features show minimal correlation with outcomes across caste groups.

Language (Severity: LOW)

Minimal bias detected in the Language dimension. The fairness score of 1.00 indicates the model treats demographic variants largely equally, with only 0.1% average difference. The Disparate Impact Ratio of 1.00 meets the 0.80 threshold.

Root Cause: Negligible language-based performance gap. The model handles linguistic variation adequately.

Region (Severity: LOW)

Minimal bias detected in the Region dimension. The fairness score of 0.88 indicates the model treats demographic variants largely equally, with only 7.2% average difference. The Disparate Impact Ratio of 1.12 meets the 0.80 threshold.

Root Cause: Minor regional variation within acceptable bounds. Geographic features have minimal impact on predictions.

Mitigation Recommendations